

Operational Intelligence Executive Summary

One-page shareable summary for executives, architects, and engineering leaders.

Definition

Operational Intelligence is the reasoning layer between enterprise telemetry and human decision.

Why it exists

Enterprises already have dashboards, logs, traces, tickets, alerts, runbooks, changes, and postmortems. The gap is turning those operational fragments into accountable judgment under uncertainty.

The ten layers

Layer	Purpose
Signal	Capture telemetry, alerts, events, changes, and operational context.
Transaction	Reconstruct customer, process, or business journeys.
Topology	Constrain reasoning with dependencies, owners, and blast radius.
Evidence	Separate observation, inference, confirmed fact, contradiction, and missing evidence.
Reasoning	Track falsifiable hypotheses and confidence movement.
Memory	Preserve reviewed operational context and replay seeds.
Evaluation	Gate retrieval, grounding, refusal, contradiction handling, and action safety.
Decision	Create reviewable options with evidence, risk, and reversibility.
Learning	Turn reviewed outcomes into memory, patterns, fixtures, and documentation.
Operator	Preserve human approval, escalation, override, and accountability.

Operating rule

Every recommendation needs supporting evidence or an explicit statement that evidence is unavailable. Every claim needs provenance. Every hypothesis must be falsifiable. Every consequential action needs approval boundaries.

What it is not

It does not replace observability, SRE, incident management, ITSM, security response, or human command. It makes the decision path inspectable.

Reviewer evidence

Question	Pass signal
Can an SRE distinguish it from observability, AIOps, and incident management?	Boundaries and overlaps are explicit.
Can two teams implement the contracts similarly?	Layer responsibilities, outputs, and states are stable.

Question	Pass signal
Does it expose contradictions and missing evidence?	Contradictions and missing evidence affect hypothesis state.
Does it preserve human approval?	Consequential action remains a decision packet, not execution.